

ROTARY SCREW AIR COMPRESSOR — MAINTENANCE & TROUBLESHOOTING MANUAL

Equipment: Oil-Injected Rotary Screw Air Compressor

Model (simulated): SIM-CMP-01 | Motor: 22 kW | Working Pressure: 7-10 bar

Document for academic/demo use — simulated non-critical equipment

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SECTION 1: EQUIPMENT OVERVIEW

The SIM-CMP-01 is a standard oil-injected rotary screw air compressor used to supply compressed air for pneumatic tools, process air, and general industrial use. Major components include the screw airend, drive motor, oil separator tank, oil cooler, air filter, and control panel.

Key components:

- Screw airend (male/female rotors)
- Drive motor and coupling/belt
- Oil separator and oil cooler
- Intake air filter
- Minimum pressure/check valve
- Safety relief valve
- Control panel with pressure/temperature sensors
- Moisture drain (automatic or manual)

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SECTION 2: SAFETY PRECAUTIONS

WARNING: Compressed air systems store significant stored energy.

All maintenance must be performed by trained personnel only, following site-specific lockout/tagout (LOTO) procedures.

Before ANY inspection or maintenance:

1. Press the emergency stop / switch off the compressor.
2. Isolate and lock out the electrical supply.
3. Fully depressurize the system — open the manual drain/bleed valve and confirm the pressure gauge reads zero before opening any part of the air system.
4. Allow the unit to cool; oil and surfaces can be hot after running.
5. Never open the oil separator tank, filters, or any pressurized component while the system shows residual pressure.
6. Relieve pressure gradually — do not vent rapidly near personnel.

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SECTION 3: NORMAL OPERATING INDICATORS

Parameter	Normal Range
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Discharge air temperature	Typically 80-100°C; alarm commonly set around 105-110°C (verify against actual unit's rated limit)
Working pressure	Within compressor's rated band (e.g. 7-10 bar for general industrial use)
Oil level	Between MIN and MAX marks on sight glass
Oil pressure differential	Within manufacturer-specified range; rising differential = filter clogging
Belt tension (belt-driven units)	Per manufacturer spec, checked by deflection test

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SECTION 4: PREVENTIVE MAINTENANCE SCHEDULE

DAILY:

- Check oil level via sight glass
- Check for unusual noise or vibration
- Drain moisture from receiver tank (if manual drain)
- Check discharge pressure and temperature on control panel

WEEKLY:

- Check for air/oil leaks around fittings and hoses
- Clean intake area of dust/debris
- Check belt tension (belt-driven units)

MONTHLY:

- Inspect/clean air intake filter; replace if heavily loaded
- Check safety relief valve operation (per manufacturer test procedure)
- Check automatic condensate drain operation

EVERY 2,000 HOURS (or per manufacturer schedule):

- Change oil filter
- Change air filter element
- Inspect drive coupling/belt condition

EVERY 4,000 HOURS (or per manufacturer schedule):

- Change compressor oil (interval varies significantly by oil type — mineral oil typically shorter interval, synthetic oil longer; always follow the specific oil manufacturer's recommendation)
- Inspect/clean oil separator element
- Full inspection of safety valve and pressure switches

ANNUALLY:

- Full inspection of airend condition (vibration/noise trending)
- Pressure vessel inspection per local regulatory requirements

- Complete electrical and control system check

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SECTION 5: COMMON FAULT INDICATIONS (ERROR CODES)

These codes are used by this project's monitoring/error-code database.

Code	Description	Severity	Typical Cause(s)
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C-E01	High discharge air temperature	High	Low oil level, clogged oil cooler, blocked ventilation, overload
C-E02	Low oil pressure/level warning	High	Oil leak, insufficient top-up, worn seals
C-E03	High discharge pressure	Medium	Faulty pressure switch, downstream valve closed, control fault
C-E04	Clogged air filter warning	Low	Dust accumulation, overdue filter change
C-E05	Excessive vibration/noise	Medium	Belt wear, coupling misalignment, airend wear
C-E06	Moisture drain fault	Low	Blocked drain valve, solenoid failure
C-E07	Motor overload trip	High	Electrical fault, mechanical binding, voltage issue
C-E08	Safety valve discharge / over-pressure	High	Faulty pressure control, blocked downstream line

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SECTION 6: TROUBLESHOOTING PROCEDURES

C-E01 — High Discharge Air Temperature

1. Stop and isolate the unit; depressurize fully (Section 2).
2. Check oil level via sight glass; top up if low (correct oil type only).
3. Check oil cooler fins for dust/debris blockage; clean if needed.
4. Check ambient ventilation around the unit is not restricted.
5. If temperature remains high after these checks, escalate to a qualified technician — possible thermostatic valve or cooler fault.

C-E02 — Low Oil Pressure/Level Warning

1. Stop and isolate the unit; depressurize fully.
2. Check sight glass oil level; inspect for visible leaks at seals, fittings, and the oil separator tank.
3. If a leak is found, do not restart until repaired.
4. Top up with the manufacturer-specified oil type only — do not mix oil types.

C-E03 — High Discharge Pressure

1. Stop and isolate the unit; depressurize fully.
2. Check that downstream isolation valves are open and not blocked.
3. Check the pressure switch/transducer for correct calibration.
4. If pressure remains abnormally high after these checks, escalate.

C-E04 — Clogged Air Filter Warning

1. Stop and isolate the unit.
2. Remove and inspect the intake air filter element.
3. Clean (if reusable type) or replace per maintenance schedule.
4. Reset filter warning indicator per control panel instructions.

C-E05 — Excessive Vibration/Noise

1. Stop and isolate the unit.

2. Check belt tension and condition (belt-driven units); check coupling alignment (direct-drive units).
3. Inspect mounting bolts for looseness.
4. If vibration/noise persists after mechanical checks, escalate — possible airend bearing wear.

C-E06 — Moisture Drain Fault

1. Stop and isolate the unit; depressurize fully.
2. Inspect manual/automatic drain valve for blockage.
3. Clean or replace the drain solenoid/valve if faulty.

C-E07 — Motor Overload Trip

1. Stop and isolate the unit.
2. Check for mechanical binding in the airend or drive coupling.
3. Verify supply voltage and phase balance.
4. If no mechanical/electrical fault is found, escalate for motor and control circuit testing.

C-E08 — Safety Valve Discharge / Over-Pressure

1. Stop and isolate the unit immediately.
2. Do NOT attempt to service the safety/relief valve yourself.
3. Escalate immediately to a qualified technician — this is a critical safety event requiring professional inspection before the unit is returned to service.

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SECTION 7: SAFETY WARNING & ESCALATION

This document provides decision-support guidance only. Compressed air systems present significant stored-energy hazards. If a fault cannot be resolved using the steps above, if pressure cannot be confirmed at

zero, or if there is any doubt about safety, escalate immediately to a qualified maintenance technician. Do not attempt repairs beyond your training or facility authorization.

END OF DOCUMENT